Case Study - Shiv Singh

Task 1: Financial Reconciliation

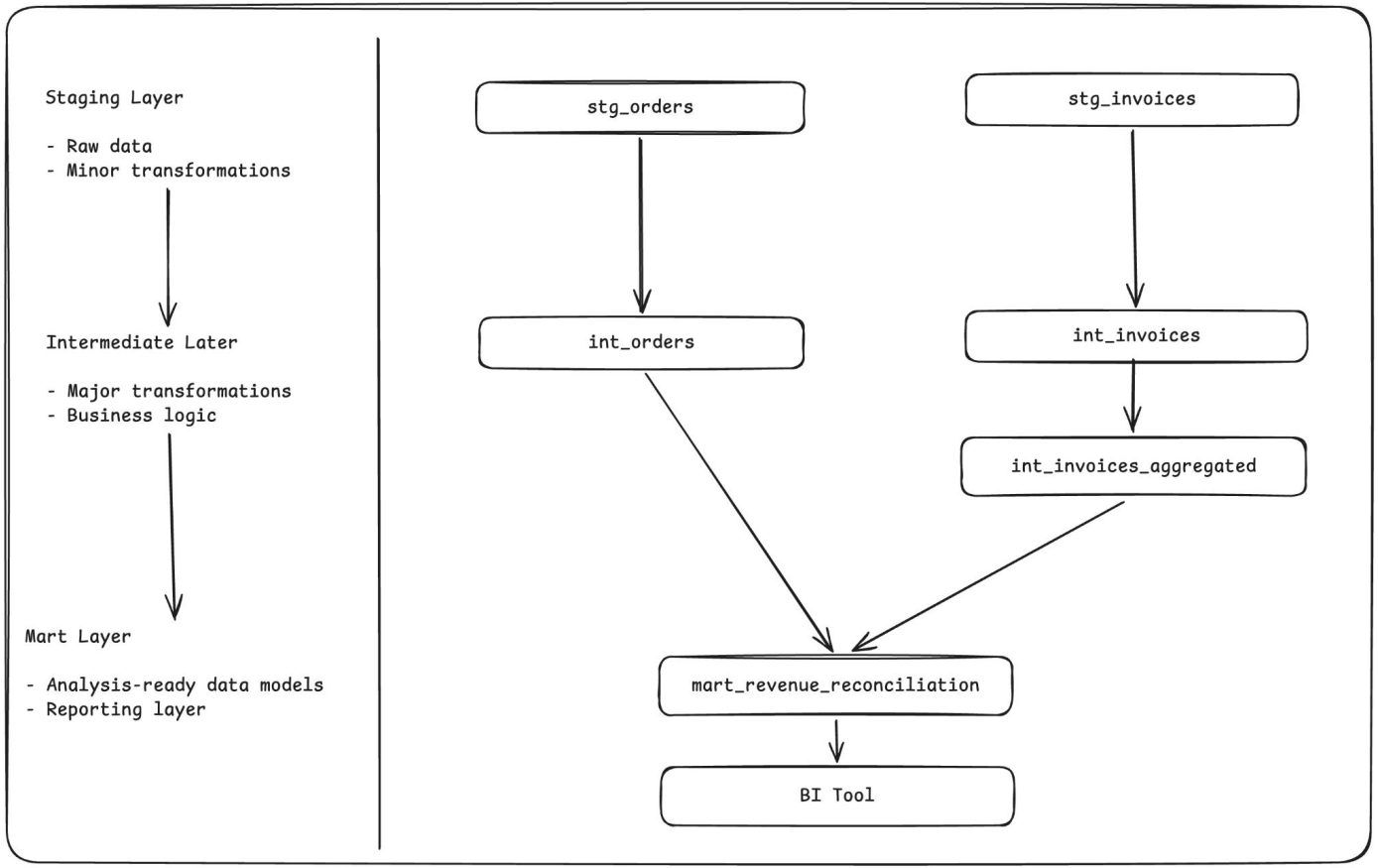
Problem Statement

Mismatch between backend order revenue and invoiced revenue, with backend showing higher values. This leads to: Inaccurate financial reporting, tax and compliance risks, poor visibility into actual cash flow. We need a scalable reconciliation process between the internal order system and invoicing system to identify, quantify, and prevent discrepancies proactively via an automated daily reconciliation model.

Investigative framework

- Question 1: How is revenue recognised in each system? What is the date anchor? (definition)
- Question 2: Do the two systems cover the same orders? (coverage)
- Question 3: Where the orders match, do amounts match? (accuracy)
- Question 4: Where the amounts match, do revenue anchor dates agree? (timing)
- Question 5: Are refunds handled consistently? (completeness)
- Question 6: Are currencies consistent across both systems? (FX conversions)

Data Model (dbt)



Key Assumptions

	Orders	Invoices
Definition	Revenue = Order Amount - Refund Amount	Revenue = Invoices Amount - Refund Amount
Anchor	Date Anchor: Order Paid Date	Date Anchor: Invoice Refund Date if available else Invoice Date
Filters	Only orders with available Order Paid Date contributes to Revenue	All the invoices along with orphan refund invoices

Net Financial Impact: €19.5k (0.9% of Total Order Net Revenue)

- **72% (€16k) of the discrepancy is driven by refund invoices**, primarily due to refunds recorded in Jan 2026 of historical orders that have no corresponding entries in Internal Orders system.
- **An additional 5% of the discrepancy is due to refund amount mismatches** between the two systems, impacting 32 orders.

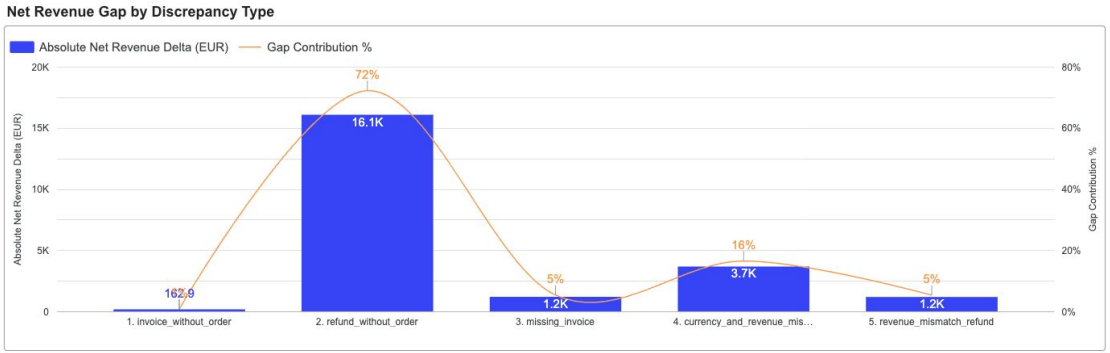


fig. 1: Net Revenue Gap by Types

- **Next biggest contributor (16%) is currency mismatch**, where CHF-denominated orders are invoiced in EUR at the same nominal value, resulting in under-invoicing. This occurs only for orders created in CHF, as all invoices are processed in EUR.
- **42 orders (5% of discrepancy) recognized as revenue in Jan 2026 have no corresponding invoices**, with the majority of these cases (81%) linked to single purchase product types (*"occ_once"*).
- 211 orders **paid** in **Feb 2026** are excluded from the analysis, as they fall outside the Jan 2026 scope of this review.

Process Improvements and Monitoring

- **Capture refund events in Orders system:**
  - Record refund events with a *refund_date* in the Orders system, enabling net revenue to be recognized in the correct period and aligned with revenue logic from Invoice system.

- **Missing Orders and Invoices:**
  - Establish automated *red flag reports* to proactively identify invoices missing orders and escalate discrepancies to Backend Engineering for proactive debugging and resolution.
  - Also flag cases where *paid orders do not have an invoice within 2 days*, with a focus on single-purchase (*occ_once*) products.

- **Currency consistency validation:**
  - Enforce rule: *invoice currency must match order currency*, with automated alerts for mismatches. Or update *order currency* to be same as *invoice currency*.

- **Ongoing reconciliation monitoring:**
  - Track a reconciliation KPI and maintain discrepancy levels below **0.1%**.

Detailed Dashboard along with Red flags report - Looker Studio

QUESTIONS

Task 1: Financial Reconciliation

Problem Statement

The operations team is managing a complex clinical fulfillment pipeline with no structured visibility into performance, decisions are driven by gut feel and ad-hoc Excel exports while customer complaints are rising. This leads to: undetected bottlenecks in case resolution, unbalanced team workload, and reactive rather than preventive ops management. We need a metrics framework and a scalable data model that gives the ops team real-time visibility into fulfillment health, case throughput, and team capacity and efficiency, proactively surfacing issues before they reach the customer.

Investigative framework

- Question 1: Are orders being fulfilled and revenue collected as expected? (payment pipeline health)
- Question 2: Are consultations being resolved within an acceptable time frame? (clinical pipeline throughput)
- Question 3: Where are cases getting stuck and what is causing the blockage? (bottleneck identification)
- Question 4: Are certain orders generating disproportionate ops effort through repeat contact? (quality signal)
- Question 5: Is the team's workload distributed fairly and is performance consistent across assignees? (capacity and efficiency)

Metrics Tree

NORTH STAR

On-time case resolution rate

% of consultations resolved within 72 calendar hours

DOMAIN 1 - ORDER FULFILLMENT

Is revenue being collected?

KPI 1 - Order completion rate
(Paid orders / Total Orders)

KPI 2 - Partial payment rate
(Partially Paid orders / Total Orders)

KPI 3 - Payment collection lag
(Days to Payment - Median & P90)

DOMAIN 2 - CONSULTATION OPS

Is the team resolving cases?

KPI 4 - Consultation resolution rate
(Resolved Consultations / Total Consultations)

KPI 5 - Resolution time SLA
(Calendar Hours to Resolution - Median & P90)

DOMAIN 3 - FULFILLMENT QUALITY

Are cases clean or looping?

KPI 6 - Stuck queue rate
(Stuck Consultations / Total Consultations)

KPI 5 - Repeat contact rate
(Orders with >= 3 consultations / Total Orders)

DOMAIN 4 - TEAM CAPACITY

Is load distributed fairly?

KPI 7 - Assignee load
(Consultations by Type / Total Consultations)

- Total Consultations
- Resolution Rate
- Stuck Rate
- P50 Resolution time

Data Model (dbt)

Staging Layer

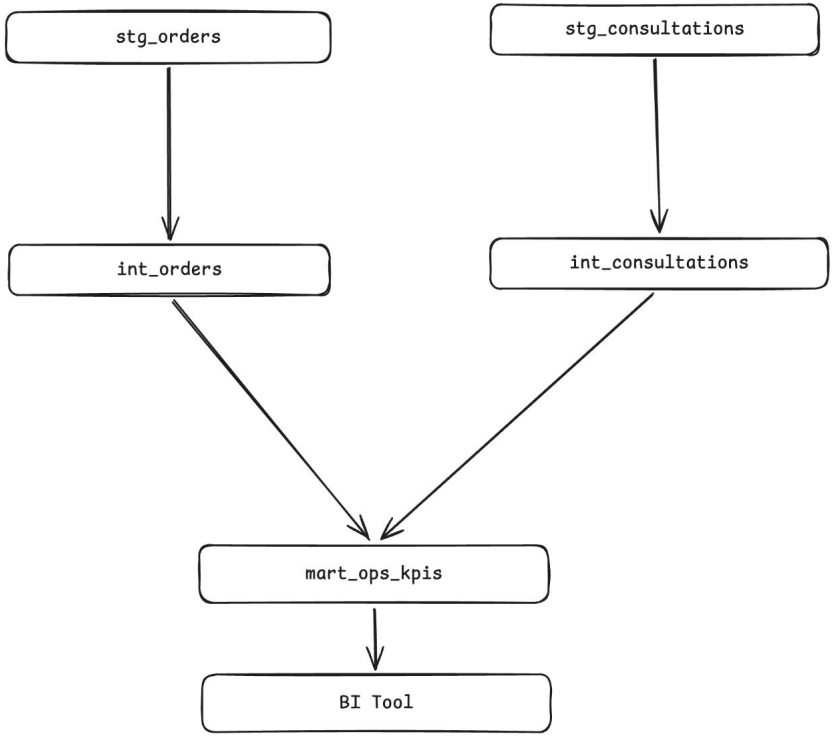
- Raw data
- Minor transformations

Intermediate Later

- Major transformations
- Business logic

Mart Layer

- Analysis-ready data models
- Reporting layer



Dashboard (Looker Studio) and Insights

- Meidan Resolution time is 12.8h (calendar hours) which sounds fine but then 90th Percentile **134h** which is alarming. That means 1 in 10 patient has to wait ~5 days for resolution.
- 27.4% of consultations are in Stuck state, of which **97%** is **payment-blocked**. This has huge implications on revenue but not necessarily an Ops failure.
- 5% of Orders have more than 3 consultations. These orders cost 3x the ops resource and have highest churn probability.

QUESTIONS